MILITARY SPECIFICATION

BELTING SYSTEM, AMMUNITION, MARK 2 MOD 0

(FOR 20MM MARK 100 SERIES AMMUNITION)

1. SCOPE

1.1 Scope. - This specification covers the procurement requirements for 20MM Ammunition Belting System Mark 2 Mod 0 referred to herein as "belting system".

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes; Wood, Wirebound.
PPP-B-601 - Boxes; Wood, Cleated-Plywood.
PPP-B-621 - Boxes; Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116 - Preservation, Methods of.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier Materials, Waterproof, Flexible.
MIL-A-140 - Adhesive, Water Resistant, Waterproof Barrier Material.
MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-C-16173 - Corrosion Preventive, Solvent Cutback, Cold-Application.
MIL-G-16908 - Grease, Bearing (for General Ordnance Use).

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STANDARDS

MILITARY

MIL-STD-8 - Dimensioning and Tolerancing.
MIL-STD-10 - Surface Roughness, Waviness, and Lay.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

BUREAU OF ORDNANCE

NAVORD OSTD 52 - Painting of Naval Ordnance Equipment.

DRAWINGS

BUREAU OF ORDNANCE

LD 477298 - Ammunition Belting System Mark 2 Mod 0 (For 20MM, T130 Ammunition and Mark 2 Link) List of Drawings, Subassemblies, Connecting Parts, Etc.
736515 - Link Handling Strip.
881289 - 20MM Ammunition Link Mk 2 Mod 0.
1445196 - 20MM Dummy Round (Machined) Mk 104 Mod 0.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 - General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

HANDBOOK H28 - Screw Thread Standards for Federal Services and Supplements Thereto.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 General. - General requirements for the manufacture of belting systems shall be in accordance with Publication OP 400. The belting system covered by this specification shall conform to all the drawings referenced in LD 477298 and to this specification and the contract or order.

3.1.1 Contractor obligation. - Means and methods of manufacture shall be determined by the contractor. The contractor is obliged to satisfy the procuring activity as to full compliance with the contract and this specification.

3.2 Materials. - The materials used in the manufacture of belting system detail parts shall be in strict accordance with the material specifications referenced on the applicable drawings.

3.3 Surface roughness and tolerances. -

3.3.1 Surface roughness. - Surface roughness shall be as specified on the applicable drawings and shall be interpreted in accordance with Standard MIL-STD-10.

3.3.2 Tolerances. - Tolerances not otherwise specified on the applicable drawings shall conform to the requirements of Standard MIL-STD-8.

3.4 Assembly. - There shall be no alteration or fitting of detail parts and subassemblies required for assembly and proper operation of belting systems. There shall be no shimming except as specified on the applicable drawings.

3.5 Interchangeability. - All parts having the same part number shall be completely interchangeable. Parts joined together by welding, brazing, or riveting shall be interchangeable as assemblies.

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3.6 Spare parts. - Spare parts furnished with assemblies shall be manufactured concurrently with like parts used in assembled belting systems.

3.7 Threads. - Unless otherwise specified on the applicable drawings, all threads shall conform to Handbook H28, and the NOT-GO gaging practice shall be as set forth in the supplement to the aforementioned handbook. The class of fit shall be as specified on the applicable drawings.

3.8 Gages. - The contractor shall provide himself with whatever gages are necessary to insure that the parts and assemblies covered by this specification will meet the dimensions and tolerances specified on the applicable drawings. If so desired, and upon request of the contractor, the Government will furnish available drawings of the applicable Government final inspection gages. This procedure will not serve to relieve the contractor of his responsibility in the design and manufacture of such other gages as may be required to satisfactorily fulfill the contract requirements.

3.9 Corrosion protection. - During the course of manufacture the surface of all parts and assembled units shall be protected against corrosion by means and methods selected by the contractor.

3.10 Painting. - Painting shall be in accordance with the applicable drawings and Standard NAVORD OSTD 52.

3.11 Identification Marking. - Identification marking of parts and assemblies not otherwise specified on the applicable drawings shall conform to the requirements of Standard MIL-STD-130.

3.12 Moisture and fungus resistance. - The requirements of Specification MIL-T-152 shall apply to nonmetallic parts and materials subject to moisture and fungus attack, except that individual parts shall be treated prior to assembly.

3.13 Construction. -

3.13.1 Power linking. - The belting system, when preset for linking operation, shall assemble links and ammunition into a continuous, uniform ammunition belt. When links and ammunition are continuously fed to the belting system, linking shall be continuous and uninterrupted. The belting system, when power operated and with the motor plate in its adjusted position for maximum operational speed, shall link ammunition at a rate of at

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least 250 rounds per minute. The belting system, with the motor plate in its adjusted position for minimum operational speed, shall link ammunition at a rate of not more than 85 rounds per minute. The belting system shall deliver ammunition belts with the ammunition in the ratios as established by the preset positions of the selector bars. Repetition of the ammunition ratios shall be uniform throughout the length of the ammunition belt.

3.13.2 Manual linking. - When the belting system is manually operated, linking shall be identical in all respects to power linking as specified in 3.13.1 except that the rate of linking shall necessarily be reduced.

3.13.3 Delinking. - The belting system, when preset for delinking operation, shall delink ammunition belts without interruption. Delinked ammunition and links shall clear the belting system so as not to foul the following ammunition and links. The belting system, when power operated and with the motor plate in its adjusted position for maximum operational speed, shall delink ammunition belts at a rate of at least 250 rounds per minute. The belting system, with the motor plate in its adjusted position for minimum operational speed, shall delink ammunition belts at a rate of not more than 85 rounds per minute. When the belting system is manually operated, delinking shall be identical in all respects to power delinking except that the rate of delinking shall necessarily be reduced.

3.14 Government loaned property. - Unless otherwise specified in the contract or order (see 6.2), property furnished by the Government on loan shall consist of ninety 20MM links Mark 2 Mod 0 conforming to Drawing 881289, ten link strips conforming to Drawing 736515, and eighty-one 20MM T130 series dummy ammunition rounds conforming to Drawing 1445196. Responsibility for loss of or damage to Government property furnished to the contractor in good condition shall rest with the contractor. The contractor shall not machine, file, grind, scrape, or otherwise modify such property except with the prior approval of the procuring activity. Each ammunition round shall be marked for group identification. The contractor shall divide the rounds into three equal groups and paint, mark, or in some way identify each group differently so that the round ratios occurring in any ammunition belt may be readily identified. The markings used shall be on the projectile end of the round.

3.15 Lubrication. - Each belting system shall be lubricated in accordance with the applicable lubrication charts.

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3.16 Performance. - When submitted to the tests of 4.4, belting systems shall function properly in all respects in accordance with the applicable drawings and this specification. Evidence of any condition that indicates a tendency to gall or bind, or adversely affects rapidity and smoothness of function, or causes undue wear, or indicates the presence of faulty material, defects in construction, or inferior workmanship shall be cause for rejection.

3.17 Workmanship. - Workmanship shall be of such quality as to produce belting systems of uniform high quality which will meet all requirements of the applicable drawings and this specification.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Lot. -

4.1.1 Belting system. - A belting system lot shall consist of all assembled belting systems manufactured by the same process and submitted for Government acceptance at any one time.

4.1.2 Parts. - A parts lot shall consist of all like parts, including spare parts, manufactured by the same process and submitted for Government acceptance at any one time.

4.2 Sampling. -

4.2.1 Belting systems. - Each belting system of any lot presented for Government acceptance under this specification shall be subjected to all inspections specified herein and the acceptance tests of 4.4.

4.2.2 Parts. - Sampling for lot acceptance of parts shall be in accordance with the procedures of Standard MIL-STD-105.

4.3 Inspection. -

4.3.1 General. - General inspection and test procedures shall be in accordance with the provisions of Publication OP 400 and this specification.

4.3.2 Visual and dimensional. - Each belting system as specified in 4.2.1 shall be visually inspected for completeness of manufacture, freedom from defects, proper identification, and workmanship. Dimensional and other nondestructive inspections shall be made to determine compliance with applicable drawings and specifications.

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4.3.3 Parts. - Parts shall be inspected in accordance with the procedures of 4.3.2, such parts to be selected in accordance with 4.2.2.

4.3.4 Preservation, packaging, packing, and marking. - Sample belting system and parts containers shall be examined to determine whether or not preservation, packaging, packing, and marking are in accordance with section 5.

4.4 Tests. -

4.4.1 Preliminary operation. - Each belting system shall be lubricated in accordance with 3.15 prior to conducting the tests. Before conducting the linking and delinking tests the belting system shall be operated first manually and then by power without ammunition and links with the inner shift lever first set at "LINK" and then at "DELINK" to visually and audibly determine that all parts and units function properly. If the above test indicates satisfactory operation of the belting system, the following linking and delinking tests shall be performed.

4.4.2 Power linking. - Power linking tests shall be performed to determine conformance to 3.13.1. Five strings of links, each string containing fifteen links, shall be placed in successive link carriers and a total of 25 dummy rounds shall be placed in each of the three ammunition chutes during each belting test run. Only rounds with identical marking (see 3.14) shall be placed in any one chute. Set the selector bars for any ratio of linking, starting with the number one selector bar, and link a belt of 75 rounds. At least six test runs, as described above, shall be made using a different setting of the selector bars for each run. Included in the six test runs shall be at least one run at each of the specified minimum and maximum belting system operational speeds. The necessity for any additional test runs shall be determined by the Government inspector. At the end of each test run the ammunition belt as delivered by the belting system shall be examined for the specified requirements.

4.4.3 Manual linking. - Seventy-five rounds of dummy ammunition shall be linked into a belt by manual operation of the belting system to determine conformance to 3.13.2. Only one setting of the selector bars shall be required. The belt as delivered by the belting system shall be examined for the specified requirements.

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4.4.4 Delinking. - The belting system shall be tested to determine conformance to 3.13.3. Power delinking of the 75 round belts for the successive linking tests specified in 4.4.2 shall serve as required delinking tests. One belt of ammunition shall be delinked by manual operation of the belting system, the length of the belt to be delinked to be determined by the Government inspector.

4.4.5 Induced failures. - In the event of failure during any test, it shall be determined whether or not such failure was induced by damaged or defective rounds, links, or link strips. Any such induced failure shall not be cause for rejection of the belting system.

5. PREPARATION FOR DELIVERY

5.1 General. - Belting systems shall be preserved and packed in accordance with one of the following levels as specified in the contract or order (see 6.2).

5.2 Preservation. -

5.2.1 Level A. -

5.2.1.1 Cleaning. - All exterior metal surfaces shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.2.1.2 Drying. - After cleaning, any drying required shall be in accordance with procedure D-1 or D-4 of Specification MIL-P-116.

5.2.1.3 Preservation procedure. - All exterior unpainted steel surfaces shall be coated with bearing grease conforming to Specification MIL-G-16908 and wrapped in accordance with method 1 of Specification MIL-P-116. The belting system shall be lubricated in accordance with 3.15 with the exception that the hookup unit housing and the selecting and linking unit base gear box shall be drained, flushed, and filled with corrosion preventive compound grade 2 conforming to Specification MIL-C-16173.

5.2.2 Level C. -

5.2.2.1 Cleaning and drying. - Cleaning and drying of belting systems for level C shall conform to 5.2.1.1 and 5.2.1.2 respectively.

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5.2.2.2 Preservation procedures. - Level "C" preservation of belting systems shall conform to 5.2.1.3 except that the hookup unit housing and the selecting and linking unit base gear box shall not be drained of its service lubricant.

5.3 Packing. - The ammunition chutes, the chute supports, the output ramp, and the link supply tray shall be disassembled from the belting system and packed in one shipping container and the remaining assembled components shall be packed in another container.

5.3.1 Level A. - Belting system components as specified in 5.3 shall be packed in an overseas type box conforming to Specifications PPP-B-585, PPP-B-601, or PPP-B-621. Shipping containers shall be lined with a case liner fabricated from type L-2 material conforming to Specification JAN-P-125 and shall be completely sealed with water resistant adhesive conforming to Specification MIL-A-140. Seams and closures shall have a water resistance equal to or better than that of the body material. The interior of each container shall be provided with suitable blocking and cushioning so as to prevent shifting of and damage to items contained therein during handling and shipping. Each container shall be provided with suitable vents.

5.3.2 Level B. - Belting system components as specified in 5.3 shall be packed in a domestic type box conforming to Specifications PPP-B-585, PPP-B-601, or PPP-B-621. The interior of each container shall be provided with suitable blocking and cushioning so as to prevent shifting and damage to items contained therein during handling and shipping. Each container shall be provided with suitable vents.

5.3.3 Level C. - Belting systems shall be packed or otherwise prepared for shipment to insure carrier acceptance and to insure safe delivery to destination at the lowest applicable rate. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.4 Spare parts. - Unless otherwise specified in the contract or order (see 6.2), spare parts shall be preserved, packaged, and packed in accordance with the plan prepared in conjunction with the NAVORD list and furnished with the contract.

5.5 Marking. - In addition to any special marking required by the contract or order (see 6.2), shipping containers shall be marked in accordance with Standard MIL-STD-129, and with the serial number of the unit it contains.

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6. NOTES

6.1 Intended use. - The ammunition belting system covered by this specification is intended for linking 20MM T130 series ammunition with Mark 2 Mod 0 links to form a continuous belt of ammunition, the belted ammunition being either of one type or of different types occurring in the belt in repeating cycles of a preselected ratio. The belting system is also intended to delink existing ammunition belts composed of ammunition and links specified above. The belting system is intended for shore or shipboard use.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Government loaned property, if applicable (see 3.14).
(c) Level of preservation and packing required (see 5.1).
(d) Preservation, packaging, and packing of spare parts if different from 5.4.
(e) Special marking when required (see 5.5).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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